

1. Let $(V, \|\cdot\|_V)$ be a normed space and let C be a nonempty subset of V . Let $f : V \rightarrow \mathbb{R}$ be defined by $f(x) := d(x, C)$ for every $x \in V$, where $d(x, C) := \inf_{y \in C} \|x - y\|_V$.

(a) Prove that if C is a convex subset of V , then the function f is convex.

Let $x_1, x_2 \in V$ and let $\lambda \in [0, 1]$. We want to show that

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

From the definitions of $f(x_1) = \inf_{y \in C} \|x_1 - y\|_V$ and $f(x_2) = \inf_{y \in C} \|x_2 - y\|_V$, we get that

$$\forall \varepsilon > 0, \quad \exists y_1^\varepsilon, y_2^\varepsilon \in C \text{ such that } \|x_1 - y_1^\varepsilon\|_V < f(x_1) + \varepsilon \text{ and } \|x_2 - y_2^\varepsilon\|_V < f(x_2) + \varepsilon.$$

Now, we multiply the first inequality by λ and the second by $1 - \lambda$ and get

$$\begin{aligned} \lambda \|x_1 - y_1^\varepsilon\|_V + (1 - \lambda) \|x_2 - y_2^\varepsilon\|_V &< \lambda(f(x_1) + \varepsilon) + (1 - \lambda)(f(x_2) + \varepsilon) \\ &= \lambda f(x_1) + (1 - \lambda)f(x_2) + \varepsilon. \quad (*) \end{aligned}$$

Now, we want the term $\lambda x_1 + (1 - \lambda)x_2$ to appear from the left hand-side in the inequality above. To this aim, we use the triangle inequality and the scalar multiplication properties of the norm $\|\cdot\|_V$ to obtain that

$$\begin{aligned} \|\lambda(x_1 - y_1^\varepsilon) + (1 - \lambda)(x_2 - y_2^\varepsilon)\|_V &\leq \|\lambda(x_1 - y_1^\varepsilon)\|_V + \|(1 - \lambda)(x_2 - y_2^\varepsilon)\|_V \\ &= \lambda \|x_1 - y_1^\varepsilon\|_V + (1 - \lambda) \|x_2 - y_2^\varepsilon\|_V. \quad (**) \end{aligned}$$

Thus, we get from $(*)$ and $(**)$ that

$$\|\lambda x_1 + (1 - \lambda)x_2 - (\lambda y_1^\varepsilon + (1 - \lambda)y_2^\varepsilon)\|_V < \lambda f(x_1) + (1 - \lambda)f(x_2) + \varepsilon.$$

Now, notice that $\lambda y_1^\varepsilon + (1 - \lambda)y_2^\varepsilon \in C$ (Why?)

Now, from the definition of $f(\lambda x_1 + (1 - \lambda)x_2)$ as infimum, we obtain that

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \|\lambda x_1 + (1 - \lambda)x_2 - (\lambda y_1^\varepsilon + (1 - \lambda)y_2^\varepsilon)\|_V < \lambda f(x_1) + (1 - \lambda)f(x_2) + \varepsilon.$$

So, we obtain

$$f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2) + \varepsilon, \quad \forall \varepsilon.$$

Therefore, letting $\varepsilon \rightarrow 0^+$ we get

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

- (b) If $C \subset V$ is closed and the function f is convex, then prove that C is a convex subset of V .

Show that every subset $K_t := \{x \in V: f(x) \leq t\}$ is convex for every $t \geq 0$. Now, for $t = 0$ we find that

$$K_0 = \{x \in V: f(x) \leq 0\} = \overline{C} = C$$

Therefore, C is convex, which is what we wanted to prove.

2. Let $(V, +, \cdot)$ be a vector space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$ and let Ω be a convex open subset of V . Let $f : \Omega \rightarrow \mathbb{R}$ be a Fréchet differentiable function. Show that f is convex if and only if

$$(Df(u) - Df(v))(u - v) \geq 0 \quad \forall u, v \in \Omega. \quad (*)$$

(Hint: Show that $(*)$ is equivalent to $f(y) \geq f(x) + Df(x)(y - x) \quad \forall x, y \in \Omega$ and use the theorem on convexity and differentiability in the lecture notes).

Following the hint, let us show that $(*)$ is equivalent to $f(y) \geq f(x) + Df(x)(y - x) \quad \forall x, y \in \Omega$

$\Rightarrow$ Consider the function $\phi : [0, 1] \rightarrow \mathbb{R}$ defined by

$$\phi(t) := f(x + t(y - x)) - tDf(x)(y - x) \quad \forall t \in [0, 1].$$

Notice f is well defined since $x + t(y - x) = ty + (1 - t)x \in \Omega$ (because Ω is convex). Also, the function ϕ is differentiable and by the Chain Rule,

$$\phi'(t) = Df(x + t(y - x))(y - x) - Df(x)(y - x) = (Df(x + t(y - x)) - Df(x))(y - x).$$

By taking $u = x + t(y - x)$ and $v = x$ in $(*)$ we find that

$$[Df(x + t(y - x)) - Df(x)](y - x) \geq 0 \quad \underbrace{\iff}_{\text{dividing by } t \in (0, 1]} \quad (Df(x + t(y - x)) - Df(x))(y - x) \geq 0.$$

So, $\phi'(t) \geq 0$ for every $t \in (0, 1]$ and hence $\phi(1) \geq \phi(0)$, that is,

$$f(y) - Df(x)(y - x) \geq f(x) \quad \Leftrightarrow \quad f(y) \geq f(x) + Df(x)(y - x).$$

$\Leftarrow$ Now, let assume that $f(y) \geq f(x) + Df(x)(y - x) \quad \forall x, y \in \Omega$ and let us show that $(*)$ holds.

We write

$$\begin{cases} f(v) \geq f(u) + Df(u)(v - u), \\ f(u) \geq f(v) + Df(v)(u - v) \end{cases}$$

and add-up to get

$$\begin{aligned} 0 &\geq Df(u)(v - u) + Df(v)(u - v) \Leftrightarrow 0 \geq -(Df(u) - Df(v))(u - v) \\ &\Leftrightarrow (Df(u) - Df(v))(u - v) \geq 0. \end{aligned}$$

Now, having shown the equivalence above and also having shown in the lectures that f is convex if and only if $f(y) \geq f(x) + Df(x)(y - x) \quad \forall x, y \in \Omega$, we obtain that f is convex if and only if $(*)$ holds.

3. Consider the function $F(x, y) := x^3 + y^3 - 4x^2y + 2$

- (a) Show that the equation $F(x, y) = 0$ defines around the point $(1, 1)$ a unique function $y = f(x)$.

The function F is of class $\mathcal{C}^\infty$.

$$F(1, 1) = 1 + 1 - 4 + 2 = 0, \quad F_y(x, y) = 3y^2 - 4x^2 \Rightarrow F_y(1, 1) = 3 - 4 = -1 \neq 0.$$

Hence, by the implicit function theorem (IFT), there exist $\delta > 0$, $\beta > 0$ and a unique function $f : (1 - \delta, 1 + \delta) \rightarrow (1 - \beta, 1 + \beta)$ of class $\mathcal{C}^\infty$ such that

$$\forall (x, y) \in (1 - \delta, 1 + \delta) \times (1 - \beta, 1 + \beta), \quad F(x, y) = 0 \Leftrightarrow y = f(x).$$

This is the format to provide as solution if such a question appears in the exam paper!

- (b) Find a cartesian equation of the tangent line to the graph of f at the point $(1, 1)$.
The tangent line to the graph of f at the point $(1, 1)$ has cartesian equation

$$y = f(1) + f'(1)(x - 1).$$

We know from the IFT that $f(1) = 1$ and that

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))} \quad \forall x \in (1 - \delta, 1 + \delta).$$

Now,

$$F_x(x, y) = 3x^2 - 8xy \Rightarrow F_x(1, 1) = -5 \Rightarrow f'(1) = -\frac{F_x(1, 1)}{F_y(1, 1)} = -\frac{-5}{-1} = -5.$$

Therefore, the tangent line to the graph of f at the point $(1, 1)$ has cartesian equation

$$y = 1 - 5(x - 1) \Leftrightarrow \boxed{y = -5x + 6}.$$

4. Consider the function $F(x, y) := 2x^3 + y^3 - 3x^2 + xy + 1$.

- (a) Show that the equation $F(x, y) = 0$ defines around the point $(1, 0)$ a unique function $y = f(x)$.

We check as above. The function F is of class $\mathcal{C}^\infty$.

$$F(1, 0) = 0, \quad F_y(x, y) = 3y^2 + x \Rightarrow F_y(1, 0) = 1 \neq 0.$$

Hence, by the implicit function theorem (IFT), there exist $\delta > 0$, $\beta > 0$ and a unique function $f : (1 - \delta, 1 + \delta) \rightarrow (-\beta, \beta)$ of class $\mathcal{C}^\infty$ such that

$$\forall (x, y) \in (1 - \delta, 1 + \delta) \times (-\beta, \beta), \quad F(x, y) = 0 \Leftrightarrow y = f(x).$$

- (b) Show that the number $x = 1$ is a critical point for the function f and determine its nature (local maximum/minimum).

We first want to show that $f'(1) = 0$. We recall that

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))} \quad \forall x \in (1 - \delta, 1 + \delta).$$

Now,

$$F_x(x, y) = 6x^2 - 6x + y \Rightarrow F_x(1, 0) = 0 \Rightarrow f'(1) = -\frac{0}{1} = 0.$$

Therefore $x = 1$ is a critical point for the function f .

In order to determine the nature of the critical point $x = 1$, we need to find the second derivative $f''(1)$ and apply the second derivative test (from MAM1000W and other first year Maths courses). We find using the quotient and Chain rules that

$$f''(x) = -\frac{(F_{xx} + F_{xy}f'(x))F_y(x, f(x)) - F_x(x, f(x))(F_{xy} + F_{yy}f'(x))}{(F_y(x, f(x)))^2}$$

where the functions F_{xx} , F_{xy} and F_{yy} are evaluated at $(x, f(x))$.

Now, using the fact that $F_x(1, 0) = 0$ and $f'(1) = 0$, we find that

$$f''(1) = -\frac{F_{xx}(1, 0)F_y(1, 0)}{(F_y(1, 0))^2} = -\frac{F_{xx}(1, 0)}{F_y(1, 0)}.$$

Now,

$$F_{xx}(x, y) = 12x - 6 \Rightarrow f''(1) = -\frac{6}{1} = -6 < 0.$$

Hence, $x = 1$ is point of local maximum for the function f .

5. Consider the function $f(x, y) := e^{3x+y} + \sin(x - y)$

- (a) Show that the curve $\mathcal{C} := \{(x, y) \in \mathbb{R}^2 : f(x, y) = 1\}$ coincides around the point $(0, 0)$ with the graph of some function $x = \phi(y)$.

Apply the IFT to the function $F(x, y) := f(x, y) - 1$, which is of class $\mathcal{C}^\infty$. So, $F(0, 0) = 0$ and

$$F_x = 3e^{3x+y} + \cos(x - y) \Rightarrow F_x(0, 0) = 4 \neq 0.$$

Hence, by the IFT there exist $\delta > 0$, $\beta > 0$ and a unique function $\phi : (-\delta, \delta) \rightarrow (-\beta, \beta)$ of class $\mathcal{C}^\infty$ such that

$$\forall (x, y) \in (-\delta, \delta) \times (-\beta, \beta), \quad F(x, y) = 0 \Leftrightarrow x = \phi(y).$$

- (b) Write down the second order Taylor polynomial of the function $x = \phi(y)$ around the point $y = 0$.

We get

$$P(y) := \frac{\phi(0)}{0!} + \frac{\phi'(0)}{1!}(y - 0) + \frac{\phi''(0)}{2!}(y - 0)^2.$$

As before, $\phi(0) = 0$,

$$\left. \begin{aligned} \phi'(y) &= -\frac{F_y(\phi(y), y)}{F_x(\phi(y), y)} \quad \text{and} \\ \phi''(y) &= -\frac{(F_{xy}\phi'(y) + F_{yy})F_x(\phi(y), y) - F_y(\phi(y), y)(F_{xx}\phi'(y) + F_{xy})}{(F_x(\phi(y), y))^2} \\ \text{with } F_{xy}, F_{yy}, F_{xx} \text{ evaluated at } (\phi(y), y). \end{aligned} \right\}$$

So,

$$\begin{aligned} & \begin{cases} F_y = e^{3x+y} - \cos(x-y), & F_{xy} = 3e^{3x+y} + \sin(x-y), \\ F_{xx} = 9e^{3x+y} - \sin(x-y) & F_{yy} = e^{3x+y} - \sin(x-y). \end{cases} \\ \Rightarrow & \begin{cases} F_y(0,0) = 0, & F_{xy}(0,0) = 3, \\ F_{xx}(0,0) = 9, & F_{yy}(0,0) = 1. \end{cases} \Rightarrow \begin{cases} \phi'(0) = -\frac{F_y(0,0)}{F_x(0,0)} = 0, \\ \phi''(0) = -\frac{F_{yy}(0,0)}{F_x(0,0)} = -\frac{1}{4}. \end{cases} \end{aligned}$$

Thus, the the second order Taylor polynomial of the function $x = \phi(y)$ around the point $y = 0$ is

$$P(y) = -\frac{1}{8}y^2.$$

(c) Show that $y = 0$ is a point of local maximum for the function ϕ .

We found that $\phi'(0) = 0$ (which shows that $y = 0$ is a critical point to the function ϕ and that $\phi''(0) < 0$. Hence, $y = 0$ is a point of local maximum for the function ϕ .